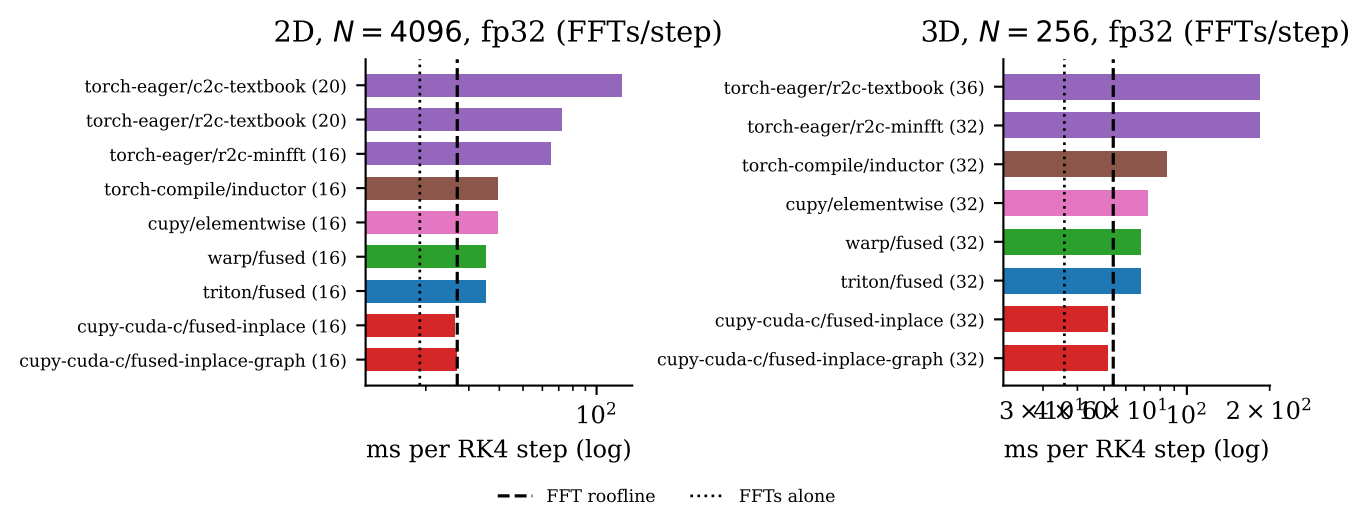
2D, $N = 4096$, fp32 (FFTs/step)
torch-eager/c2c-textbook (20)
torch-eager/r2c-textbook (20)
torch-eager/r2c-minfft (16)
torch-compile/inductor (16)
cupy/elementwise (16)
warp/fused (16)
triton/fused (16)
cupy-cuda-c/fused-inplace (16)
cupy-cuda-c/fused-inplace-graph (16)
10^2
ms per RK4 step (log)
3D, $N = 256$, fp32 (FFTs/step)
torch-eager/r2c-textbook (36)
torch-eager/r2c-minfft (32)
torch-compile/inductor (32)
cupy/elementwise (32)
warp/fused (32)
triton/fused (32)
cupy-cuda-c/fused-inplace (32)
cupy-cuda-c/fused-inplace-graph (32)
3×10^1
4×10^1
6×10^1
10^2
2×10^2
ms per RK4 step (log)
FFT roofline
FFTs alone